NeoJaundice: Low-Cost AI-Assisted Neonatal Jaundice Screening

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Hackathon: TNWISE – 2026 TANCAMs hackathon for Women
Institution: SRM Madurai College for Engineering and Technology

Abstract

Neonatal jaundice affects nearly 60 % of term and 80 % of preterm newborns and requires early detection to prevent severe hyperbilirubinemia and neurological complications. In primary healthcare settings, especially rural and low-resource environments, objective bilirubin screening devices are rarely available due to high cost and infrastructure requirements. As a result, frontline healthcare workers often rely on subjective visual assessment, which is influenced by ambient lighting conditions, observer variability, and neonatal skin tone differences. This leads to delayed identification of jaundice risk and preventable complications.

NeoJaundice proposes a low-cost AI-assisted neonatal jaundice screening approach using controlled-illumination imaging and mobile chromatic analysis. A handheld imaging module captures neonatal skin or sclera under standardized lighting conditions to ensure consistent color acquisition independent of environment. The captured image is analyzed in a mobile application using region-of-interest extraction, color normalization in YCbCr space, and chromatic feature analysis to estimate jaundice risk levels. The system provides an offline, rapid screening output suitable for primary healthcare workers.

A proof-of-concept prototype using low-cost imaging components and a demonstration dataset of neonatal skin and sclera images validates the feasibility of the approach. NeoJaundice demonstrates potential for affordable point-of-care neonatal jaundice screening in primary healthcare settings, supporting early detection and timely referral while reducing reliance on expensive hospital-based devices.

Problem & Motivation

Neonatal jaundice is one of the most common clinical conditions in newborns, affecting approximately 60 % of term and 80 % of preterm infants. While physiological jaundice is often benign, delayed recognition of elevated bilirubin levels can lead to severe hyperbilirubinemia, acute bilirubin encephalopathy, and permanent neurological damage such as kernicterus. Early identification of jaundice risk is therefore essential to enable timely monitoring and referral.

In hospital settings, bilirubin levels are typically measured using transcutaneous bilirubinometers or laboratory serum tests. However, these devices are expensive and largely unavailable in Primary Health Centres (PHCs), rural clinics, and outreach health programs. Consequently, frontline healthcare workers such as nurses and community health workers rely primarily on visual assessment of neonatal skin or scleral yellowing.

Visual assessment of jaundice is inherently subjective and influenced by multiple factors including ambient lighting conditions, observer experience, and neonatal skin pigmentation. Inconsistent lighting environments common in rural settings further reduce reliability. Additionally, subtle early-stage jaundice may not be easily recognized visually, leading to delayed detection and referral. These limitations are particularly critical in low-resource healthcare environments where specialist neonatal expertise is limited.

The combination of high device cost, lack of standardized screening tools, and variability in visual assessment creates a significant gap in early neonatal jaundice detection at the primary care level. An accessible, objective, and low-cost screening approach that can be used by frontline health workers is therefore required to improve early identification of jaundice risk outside hospital settings.

Need Statement:

A low-cost, objective, point-of-care neonatal jaundice screening tool suitable for primary healthcare environments is required to enable early detection and timely referral.

Proposed Solution

NeoJaundice proposes a low-cost, AI-assisted neonatal jaundice screening approach that enables objective color-based assessment of neonatal skin or sclera in primary healthcare settings. The system combines controlled-illumination handheld imaging with mobile chromatic analysis to provide rapid, offline jaundice risk estimation suitable for frontline health workers.

The solution consists of a handheld imaging module and a mobile AI analysis application. The handheld module captures neonatal skin or sclera images under uniform illumination using an integrated LED-diffuser optical system. Controlled illumination ensures consistent color acquisition independent of ambient lighting conditions, observer variability, and environmental differences common in rural healthcare environments. A fixed-distance spacer maintains stable imaging geometry, enabling repeatable color capture across screenings.

Captured images are transmitted to a mobile application where automated region-of-interest extraction isolates clinically relevant skin or scleral regions. The image is then normalized using luminance-independent color representation (YCbCr) to reduce brightness variation and enhance chromatic information related to bilirubin deposition. Chromatic features associated with yellow-blue balance shifts are analyzed using a lightweight classification model to estimate jaundice risk levels.

The system provides an intuitive screening output categorized as low, moderate, or high risk, along with guidance for monitoring or referral. By standardizing neonatal color capture and analysis, NeoJaundice converts subjective visual assessment into a consistent, objective screening workflow usable by non-specialist healthcare workers. The approach enables early jaundice risk identification at the point of care without reliance on expensive hospital-based bilirubin measurement devices

System Architecture

The NeoJaundice system follows a modular architecture consisting of a handheld imaging unit, a communication interface, and a mobile AI processing application. This architecture enables standardized neonatal image acquisition and automated chromatic analysis for jaundice risk estimation in primary healthcare environments.

The handheld imaging unit integrates a camera module with a controlled LED illumination system and optical diffuser to provide uniform lighting across the neonatal skin or sclera region. A fixed-distance spacer ensures stable imaging geometry and repeatable capture conditions. Embedded control electronics manage illumination timing, image acquisition, and data transmission. By standardizing illumination and capture parameters, the imaging module produces color-consistent neonatal images independent of ambient lighting and operator variability.

Captured images are transferred wirelessly or via wired interface to a mobile processing device. The mobile application performs automated region-of-interest extraction to isolate clinically relevant skin or scleral regions. Color normalization is applied using luminance-separated color space representation to minimize brightness variation and enhance chromatic information related to bilirubin deposition. Extracted chromatic features are analyzed by a lightweight classification model to estimate neonatal jaundice risk levels.

The output module presents the screening result in an intuitive risk format along with recommended monitoring or referral guidance. The modular separation of imaging and processing components enables low-cost hardware design while leveraging mobile computing capability for analysis. This architecture supports scalable deployment in primary healthcare settings and forms the basis for future clinical device development.

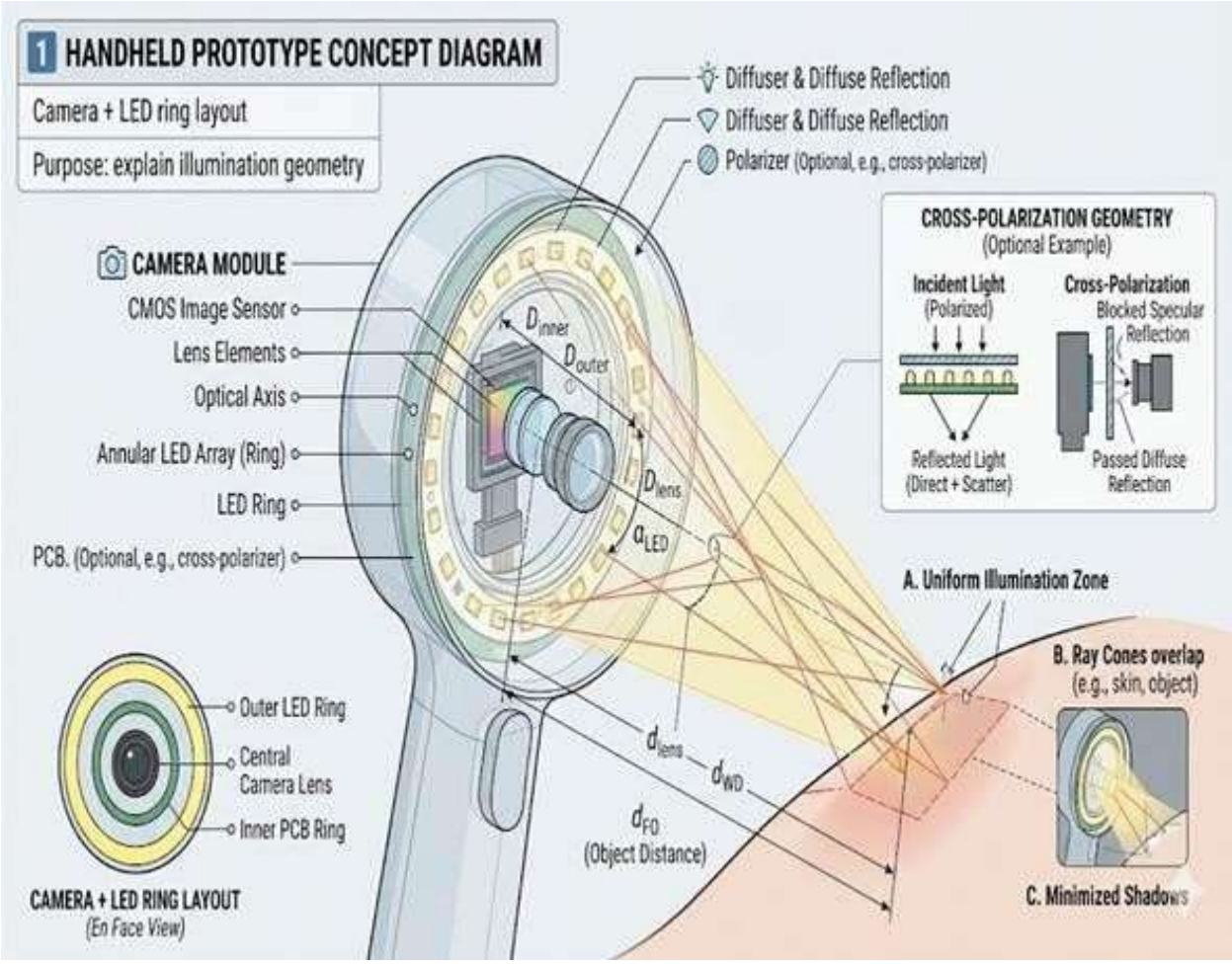
Prototype Design

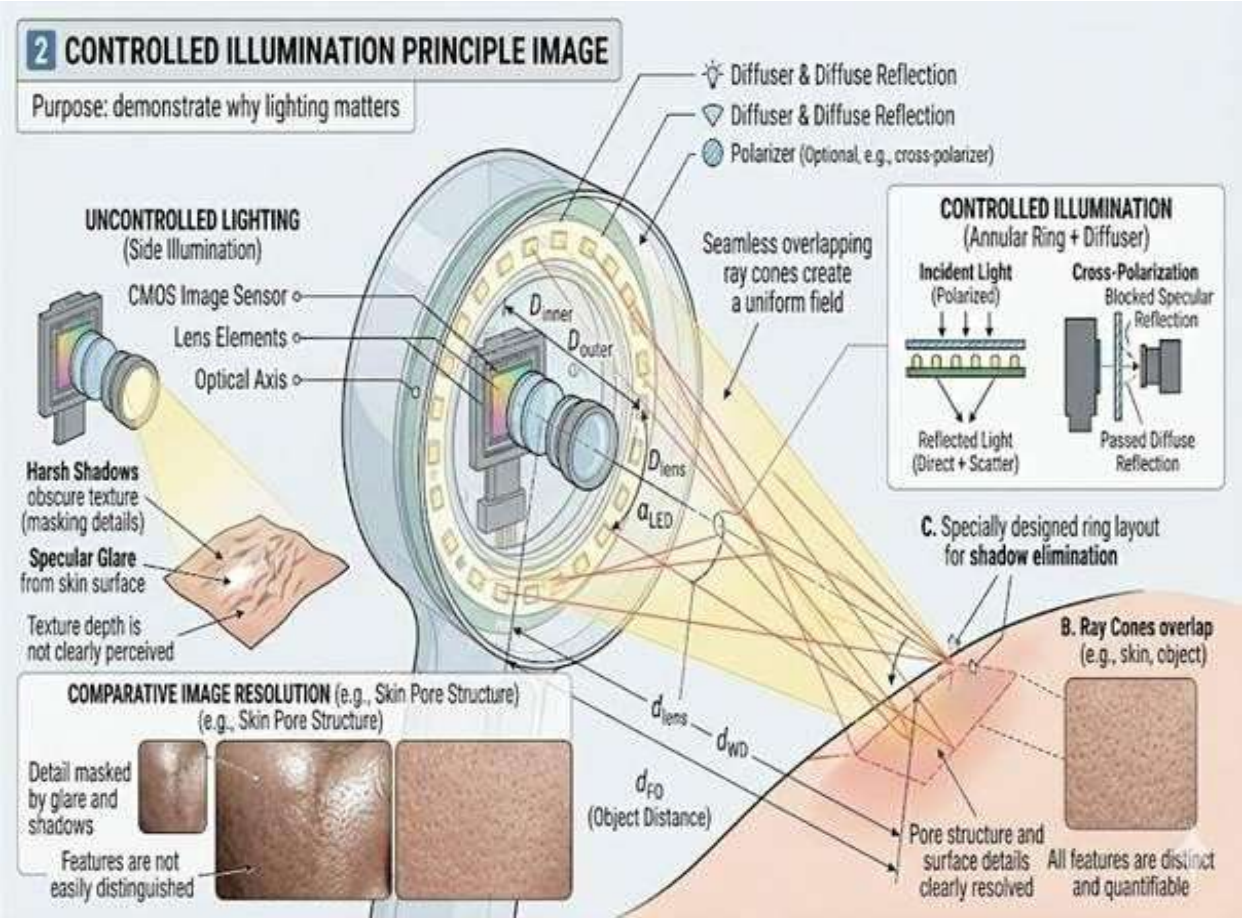
A proof-of-concept handheld imaging prototype was designed to validate the NeoJaundice controlled-illumination capture principle using low-cost, off-the-shelf components. The prototype focuses on achieving standardized neonatal color acquisition independent of ambient lighting, which is critical for reliable chromatic jaundice analysis.

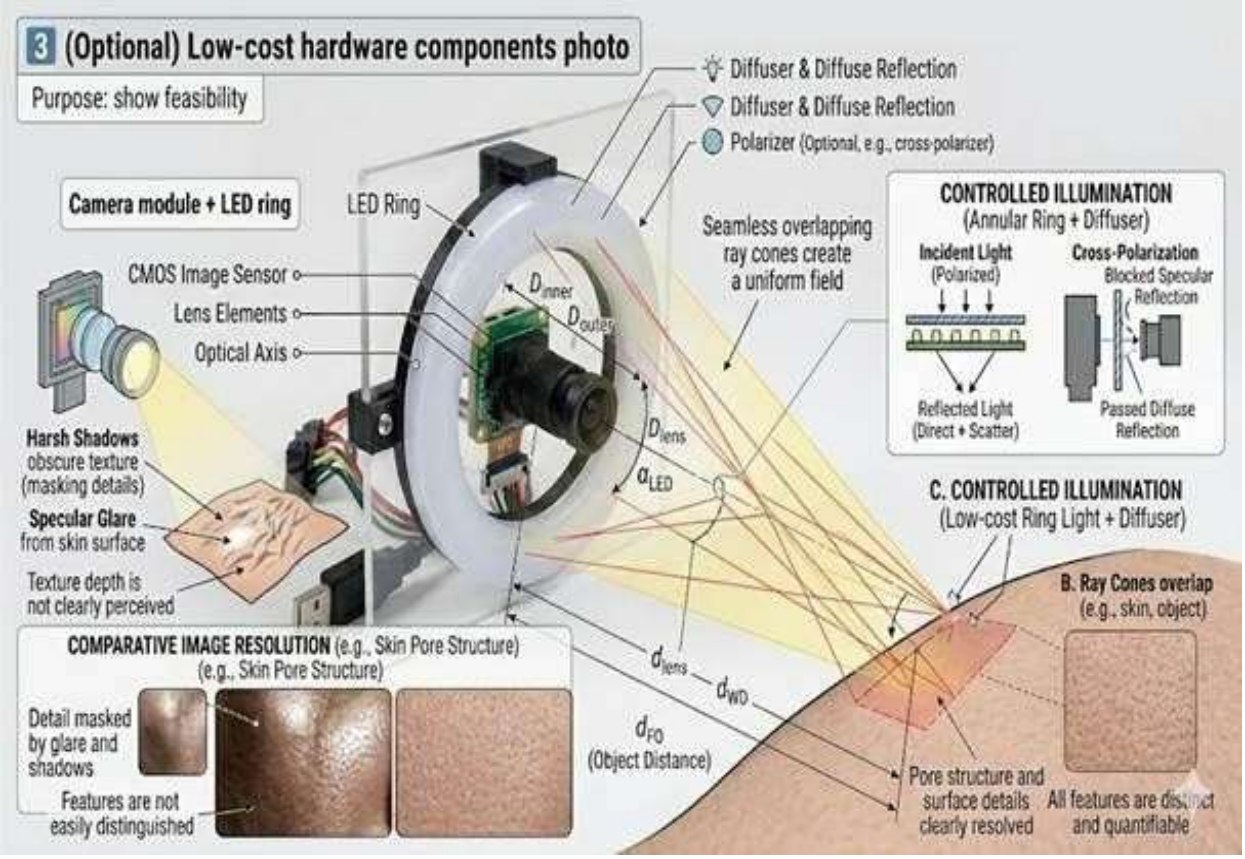
The handheld module consists of a compact camera sensor positioned concentrically within a circular white LED illumination ring. A translucent optical diffuser is placed over the LEDs to ensure uniform light distribution across the neonatal skin or sclera region. This configuration minimizes shadows and ambient light influence while preserving true color characteristics required for bilirubin-related chromatic analysis.

A fixed spacer structure maintains a constant distance between the imaging module and neonatal surface, ensuring repeatable capture geometry across screenings. Stable imaging distance and illumination consistency together enable comparable color acquisition across different environments and operators. The imaging unit is powered by a portable low-voltage supply, enabling handheld operation suitable for primary healthcare settings.

The prototype demonstrates the core principle of controlled-illumination neonatal imaging required for objective jaundice screening. By combining uniform lighting, fixed geometry, and low-cost imaging hardware, the design establishes feasibility for an affordable point-of-care screening device. The prototype architecture supports integration with a mobile AI analysis application for chromatic feature extraction and risk estimation.




2 CONTROLLED ILLUMINATION PRINCIPLE IMAGE
Purpose: demonstrate why lighting matters
Diffuser & Diffuse Reflection
Diffuser & Diffuse Reflection
Polarizer (Optional, e.g., cross-polarizer)
UNCONTROLLED LIGHTING
(Side Illumination)
CMOS Image Sensor
Lens Elements
Optical Axis
Harsh Shadows
obscure texture
(masking details)
Specular Glare
from skin surface
Texture depth is
not clearly perceived
Seamless overlapping
ray cones create
a uniform field
CONTROLLED ILLUMINATION
(Annular Ring + Diffuser)
Incident Light
(Polarized)
Cross-Polarization
Blocked Specular
Reflection
Reflected Light
(Direct + Scatter)
Passed Diffuse
Reflection
C. Specially designed ring layout
for shadow elimination
B. Ray Cones overlap
(e.g., skin, object)
COMPARATIVE IMAGE RESOLUTION (e.g., Skin Pore Structure)
(e.g., Skin Pore Structure)
Detail masked
by glare and
shadows
Features are not
easily distinguished
Pore structure and
surface details
clearly resolved
All features are distinct
and quantifiable


3 (Optional) Low-cost hardware components photo
Purpose: show feasibility
Diffuser & Diffuse Reflection
Diffuser & Diffuse Reflection
Polarizer (Optional, e.g., cross-polarizer)
Camera module + LED ring
LED Ring
CMOS Image Sensor
Lens Elements
Optical Axis
Harsh Shadows
obscure texture
(masking details)
Specular Glare
from skin surface
Texture depth is
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Seamless overlapping
ray cones create
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CONTROLLED ILLUMINATION
(Annular Ring + Diffuser)
Incident Light
(Polarized)
Cross-Polarization
Blocked Specular
Reflection
Reflected Light
(Direct + Scatter)
Passed Diffuse
Reflection
C. CONTROLLED ILLUMINATION
(Low-cost Ring Light + Diffuser)
B. Ray Cones overlap
(e.g., skin, object)
COMPARATIVE IMAGE RESOLUTION (e.g., Skin Pore Structure)
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Detail masked
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Features are not
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Pore structure and
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All features are distinct
and quantifiable

AI Methodology & Dataset

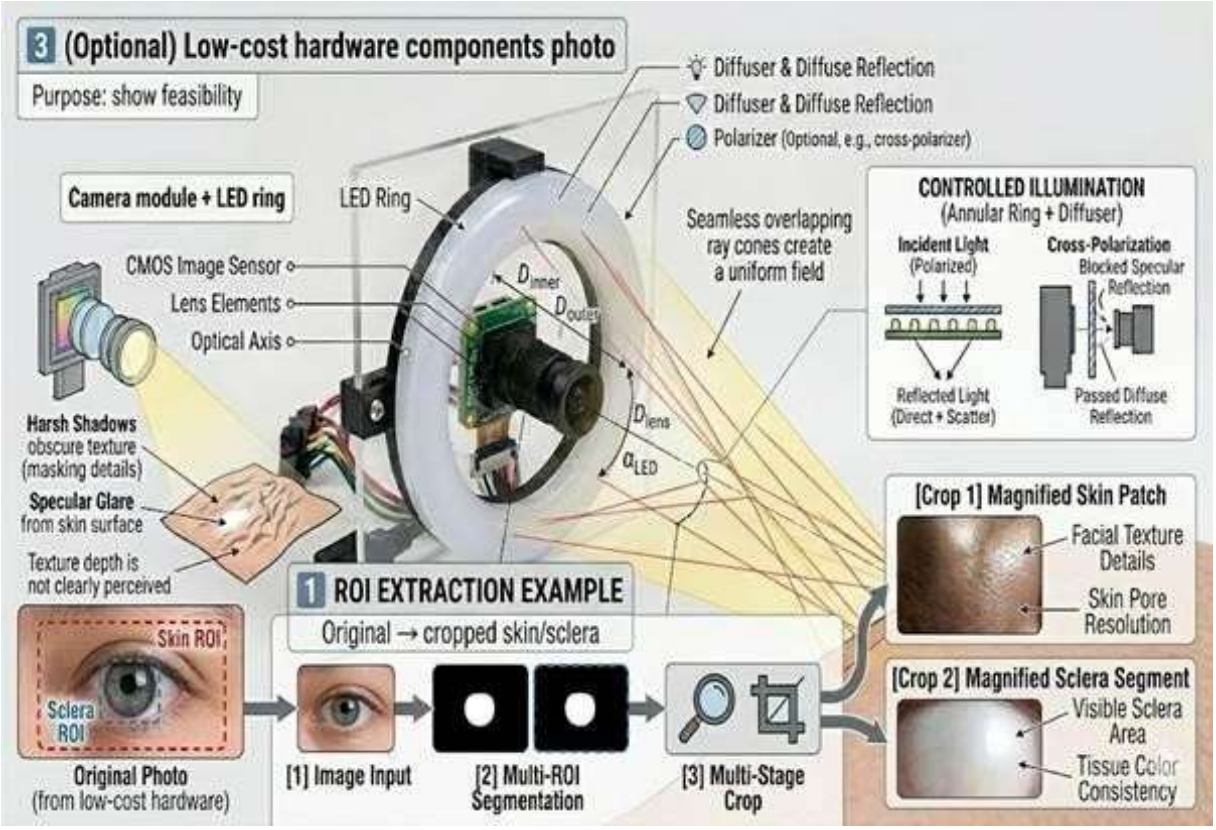
The NeoJaundice mobile application performs automated chromatic analysis of neonatal skin or sclera images to estimate jaundice risk. The AI pipeline is designed to extract bilirubin-related color features while minimizing variability caused by illumination and background differences.

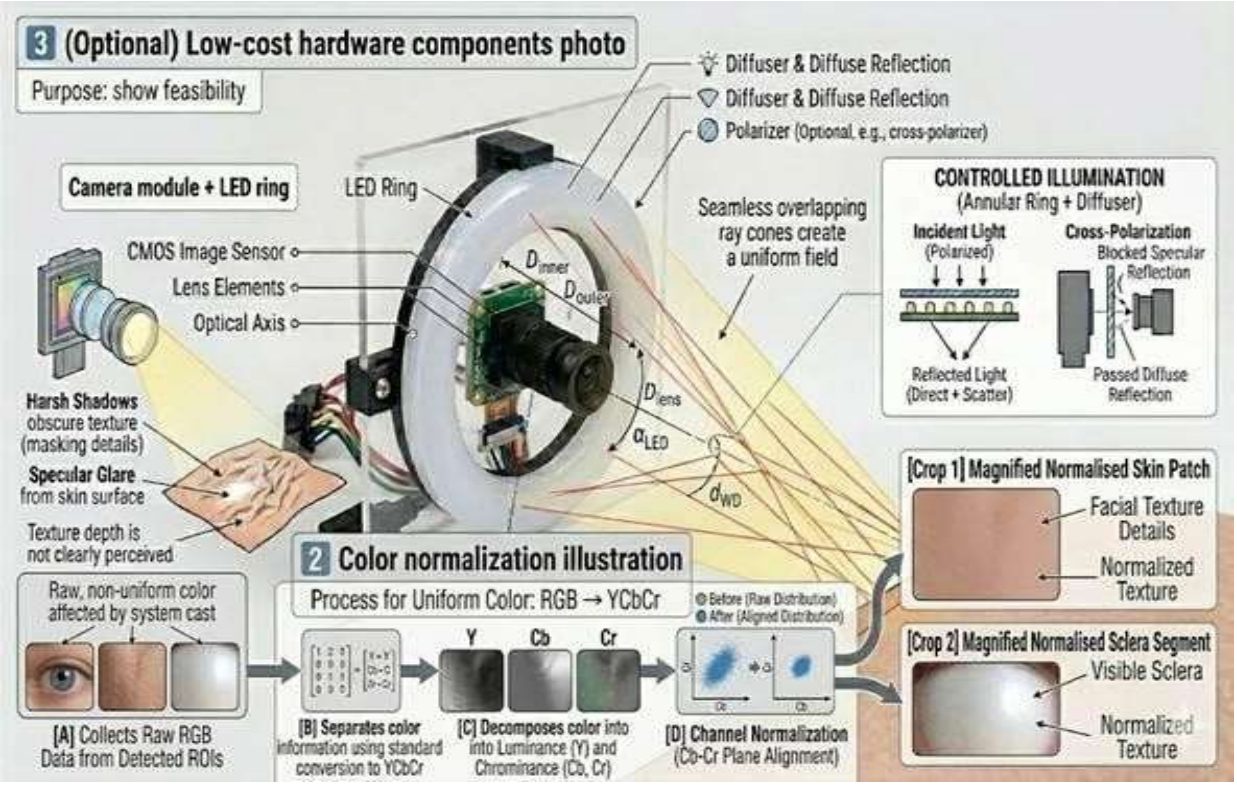
The captured neonatal image is first processed using region-of-interest (ROI) extraction to isolate clinically relevant skin or scleral areas. Background pixels and non-relevant regions are removed to ensure that subsequent analysis focuses only on tissue areas where jaundice-related chromatic changes are observable.

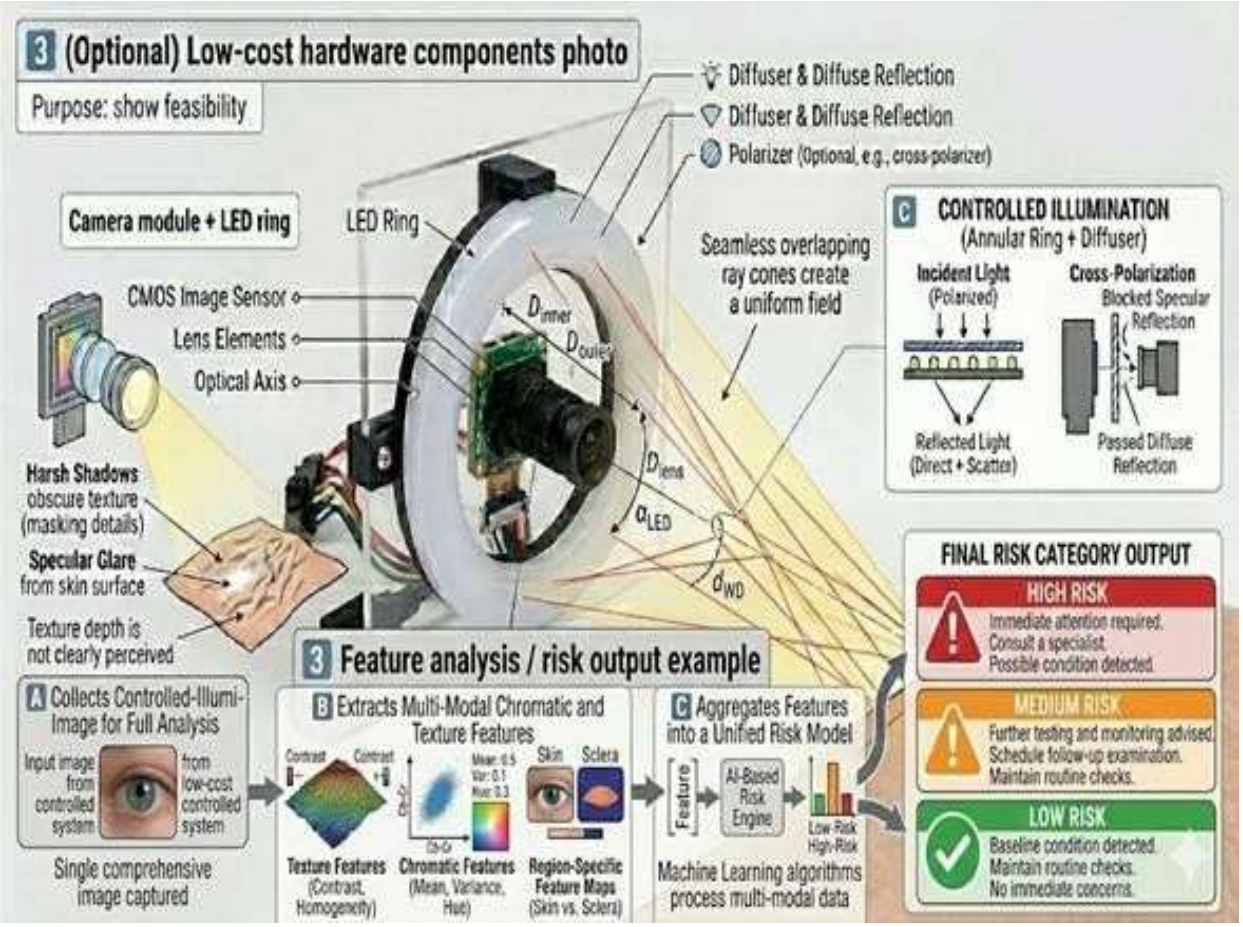
Following ROI extraction, color normalization is applied by converting the image from RGB representation to a luminance-separated color space (YCbCr). This transformation separates brightness information from chromatic components, enabling analysis of yellow-blue color balance independent of lighting intensity. Such normalization improves robustness of color-based jaundice estimation under varying illumination conditions.

Chromatic features associated with bilirubin deposition are then computed from the normalized image. These include relative chroma ratios and yellow-blue balance metrics that reflect the visual yellowing characteristic of neonatal jaundice. The extracted feature set is provided to a lightweight classification model that categorizes neonatal jaundice risk into low, moderate, or high levels suitable for screening guidance.

For hackathon demonstration, a proof-of-concept dataset consisting of neonatal skin and sclera images was used. The dataset included approximately 100–150 images representing varying skin tones and illumination conditions. Additional augmented samples were generated to simulate chromatic variation and improve model robustness. This dataset serves solely to validate the feasibility of the chromatic screening pipeline and does not represent a clinically validated training set.




3 (Optional) Low-cost hardware components photo
Purpose: show feasibility
Diffuser & Diffuse Reflection
Diffuser & Diffuse Reflection
Polarizer (Optional, e.g., cross-polarizer)
Camera module + LED ring
LED Ring
CMOS Image Sensor
Lens Elements
Optical Axis
Seamless overlapping ray cones create a uniform field
CONTROLLED ILLUMINATION
(Annular Ring + Diffuser)
Incident Light
(Polarized)
Cross-Polarization
Blocked Specular Reflection
Reflected Light
(Direct + Scatter)
Passed Diffuse Reflection
Harsh Shadows
obscure texture
(masking details)
Specular Glare
from skin surface
Texture depth is
not clearly perceived
[Crop 1] Magnified Normalised Skin Patch
Facial Texture Details
Normalized Texture
2 Color normalization illustration
Process for Uniform Color: RGB → YCbCr
Raw, non-uniform color affected by system cast
Before (Raw Distribution)
After (Aligned Distribution)
Y
Cb
Cr
[Crop 2] Magnified Normalised Sclera Segment
Visible Sclera
Normalized Texture
[A] Collects Raw RGB Data from Detected ROIs
[B] Separates color information using standard conversion to YCbCr
[C] Decomposes color into into Luminance (Y) and Chrominance (Cb, Cr)
[D] Channel Normalization (Cb-Cr Plane Alignment)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obscure texture
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Texture depth is
not clearly perceived
FINAL RISK CATEGORY OUTPUT
HIGH RISK
Immediate attention required.
Consult a specialist.
Possible condition detected.
MEDIUM RISK
Further testing and monitoring advised.
Schedule follow-up examination.
Maintain routine checks.
LOW RISK
Baseline condition detected.
Maintain routine checks.
No immediate concerns.
3 Feature analysis / risk output example
A Collects Controlled-Illumi-Image for Full Analysis
Input image from controlled system
from low-cost controlled system
Single comprehensive image captured
B Extracts Multi-Modal Chromatic and Texture Features
Contrast
Contrast
Skin
Sclera
Texture Features
(Contrast, Homogeneity)
Chromatic Features
(Mean, Variance, Hue)
Region-Specific Feature Maps
(Skin vs. Sclera)
C Aggregates Features into a Unified Risk Model
Feature
AI-Based Risk Engine
Low-Risk
High-Risk
Machine Learning algorithms process multi-modal data

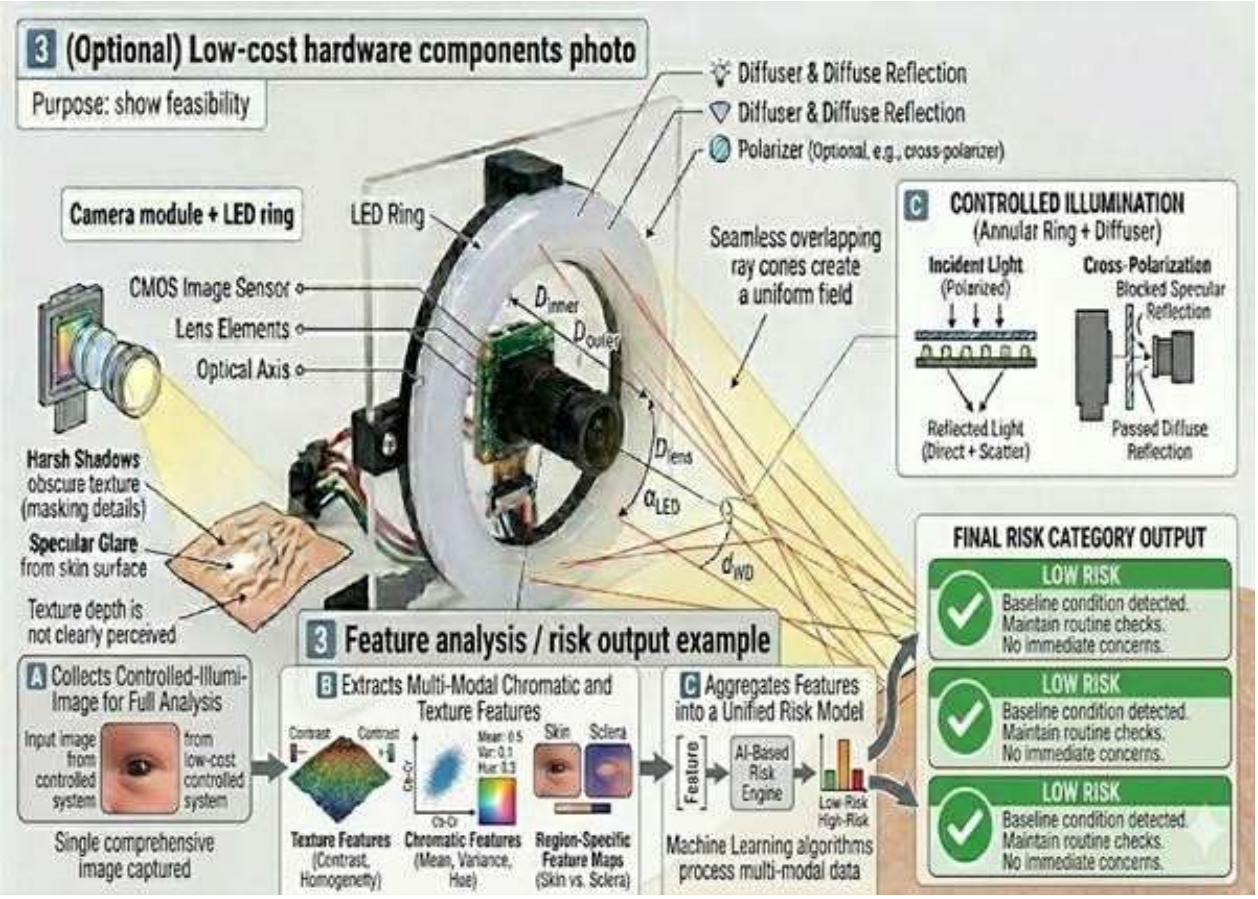
Results & Cost Analysis

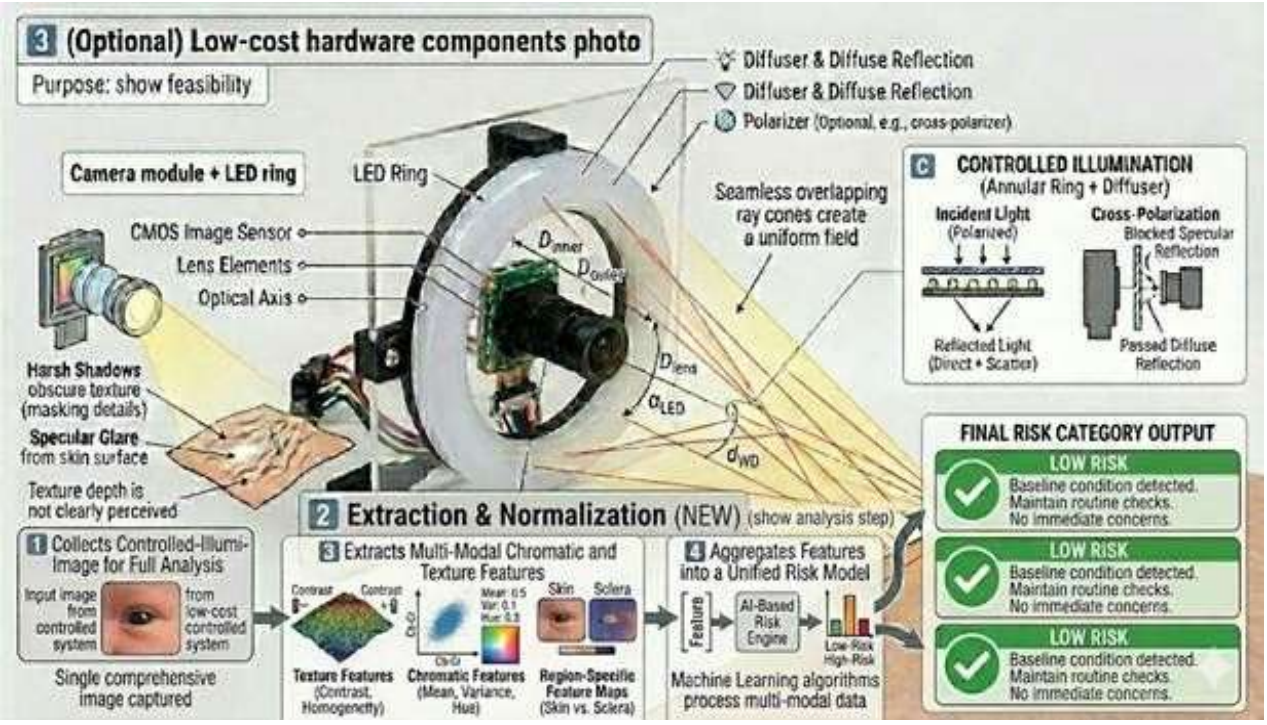
The NeoJaundice prototype workflow successfully demonstrates the feasibility of controlled-illumination neonatal imaging combined with mobile chromatic analysis for jaundice risk screening. The handheld imaging configuration enables standardized color capture independent of ambient lighting conditions, producing consistent neonatal skin or sclera images suitable for chromatic feature extraction.

The mobile AI pipeline performs automated region-of-interest extraction, color normalization in luminance-separated color space, and chromatic feature computation to estimate jaundice risk levels. The system generates intuitive screening outputs categorized as low, moderate, or high risk, demonstrating the ability to convert subjective visual assessment into an objective, repeatable screening process. These results validate the technical feasibility of low-cost point-of-care neonatal jaundice screening in primary healthcare environments.

The prototype hardware design uses low-cost, off-the-shelf imaging and illumination components, enabling an estimated handheld prototype cost of approximately ₹2,000–₹3,000. This cost level demonstrates strong affordability compared to conventional bilirubin screening devices, which are typically expensive and hospital-based. The simplified optical and electronic architecture supports potential mass manufacturing and integration into a compact handheld device.

Projected large-scale production using integrated printed circuit design and molded enclosure fabrication is expected to reduce unit cost below ₹5,000. Such cost reduction enables scalable deployment across primary healthcare centers, outreach screening programs, and rural neonatal care settings. The combination of low hardware cost and mobile AI processing establishes NeoJaundice as a financially feasible screening approach for resource-limited healthcare environments.





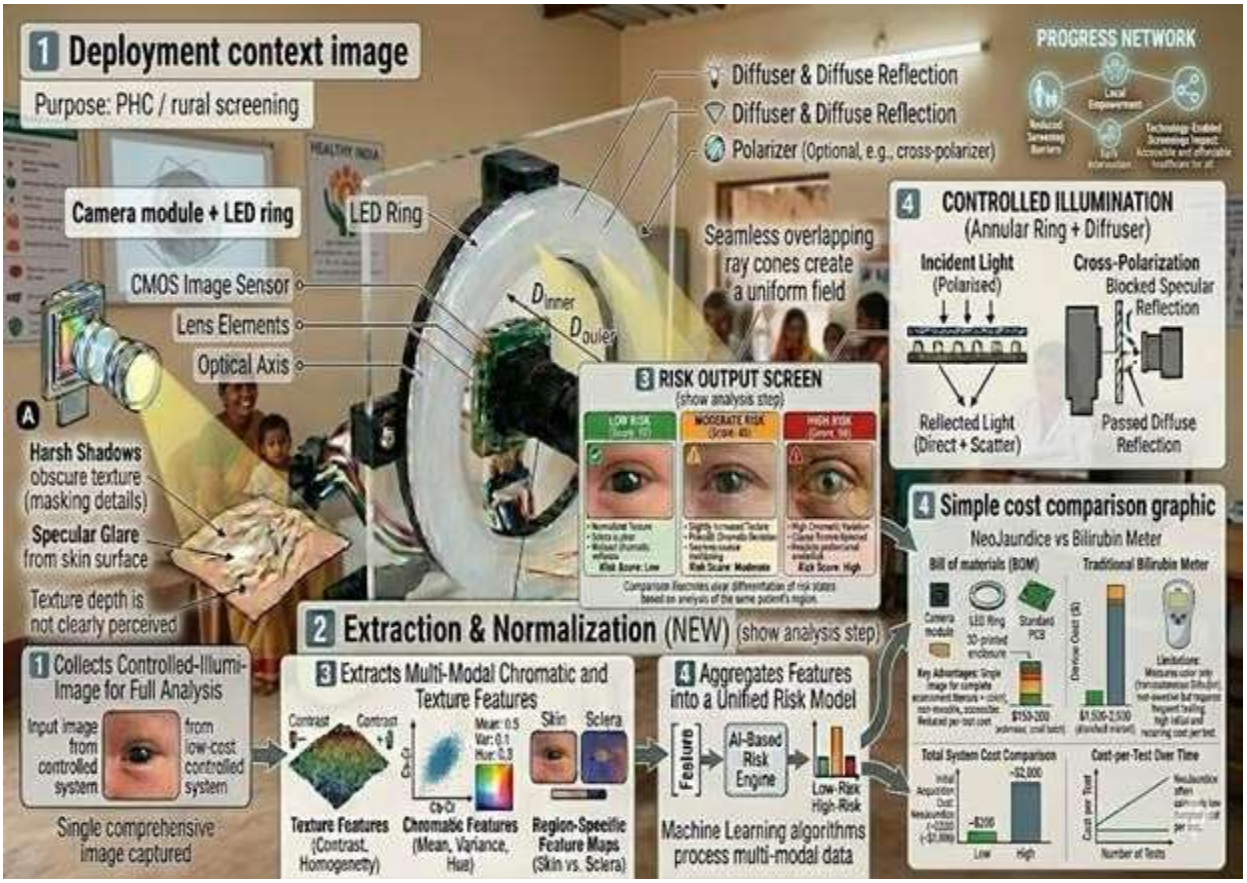
Impact, Limitations & Future Scope

The NeoJaundice system demonstrates the potential to enable early neonatal jaundice screening in primary healthcare environments where objective bilirubin assessment tools are currently unavailable. By converting subjective visual observation into standardized color-based screening, the approach supports frontline health workers in identifying jaundice risk at the point of care. The low-cost handheld design and mobile AI analysis enable accessibility in PHCs, rural clinics, and outreach screening programs. Early detection facilitated by such screening can support timely monitoring and referral, potentially reducing severe hyperbilirubinemia and preventable neonatal complications.

The proposed approach is designed for affordability and scalability. The estimated prototype cost of ₹2,000–₹3,000 and projected mass-manufactured cost below ₹5,000 indicate feasibility for large-scale deployment across resource-limited healthcare settings. Integration with mobile devices further enhances portability and usability without dependence on specialized infrastructure.

However, the current implementation has limitations typical of an early-stage prototype. The AI model was developed and demonstrated using a small proof-of-concept dataset and does not represent a clinically validated screening system. The handheld imaging prototype demonstrates controlled-illumination feasibility but has not undergone clinical testing or calibration against bilirubin measurements. Environmental, physiological, and demographic variability across neonatal populations requires larger clinical datasets for robust validation. Therefore, the system should be considered a feasibility demonstration rather than a diagnostic medical device.

Future work will focus on clinical dataset collection with paired bilirubin measurements to enable validated model training and evaluation. Optimization of illumination optics and imaging calibration will improve color accuracy and reproducibility. Integration into a compact ergonomic handheld device and usability testing in PHC environments will support real-world deployment. With regulatory validation and clinical evaluation, NeoJaundice has the potential to evolve into an affordable neonatal jaundice screening tool for primary healthcare systems.


1 Deployment context image
Purpose: PHC / rural screening
Camera module + LED ring
LED Ring
CMOS Image Sensor
Lens Elements
Optical Axis
Harsh Shadows obscure texture (masking details)
Specular Glare from skin surface
Texture depth is not clearly perceived
Diffuser & Diffuse Reflection
Diffuser & Diffuse Reflection
Polarizer (Optional, e.g., cross-polarizer)
Seamless overlapping ray cones create a uniform field
PROGRESS NETWORK
4 CONTROLLED ILLUMINATION (Annular Ring + Diffuser)
Incident Light (Polarised)
Cross-Polarization
Blocked Specular Reflection
Reflected Light (Direct + Scatter)
Passed Diffuse Reflection
3 RISK OUTPUT SCREEN (show analysis step)
4 Simple cost comparison graphic
NeoJaundice vs Bilirubin Meter
Bill of materials (BOM)
Traditional Bilirubin Meter
Total System Cost Comparison
Cost-per-Test Over Time
2 Extraction & Normalization (NEW) (show analysis step)
1 Collects Controlled-Illumi-Image for Full Analysis
Input image from controlled system
from low-cost controlled system
Single comprehensive image captured
3 Extracts Multi-Modal Chromatic and Texture Features
Contrast
Skin
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Texture Features (Contrast, Homogeneity)
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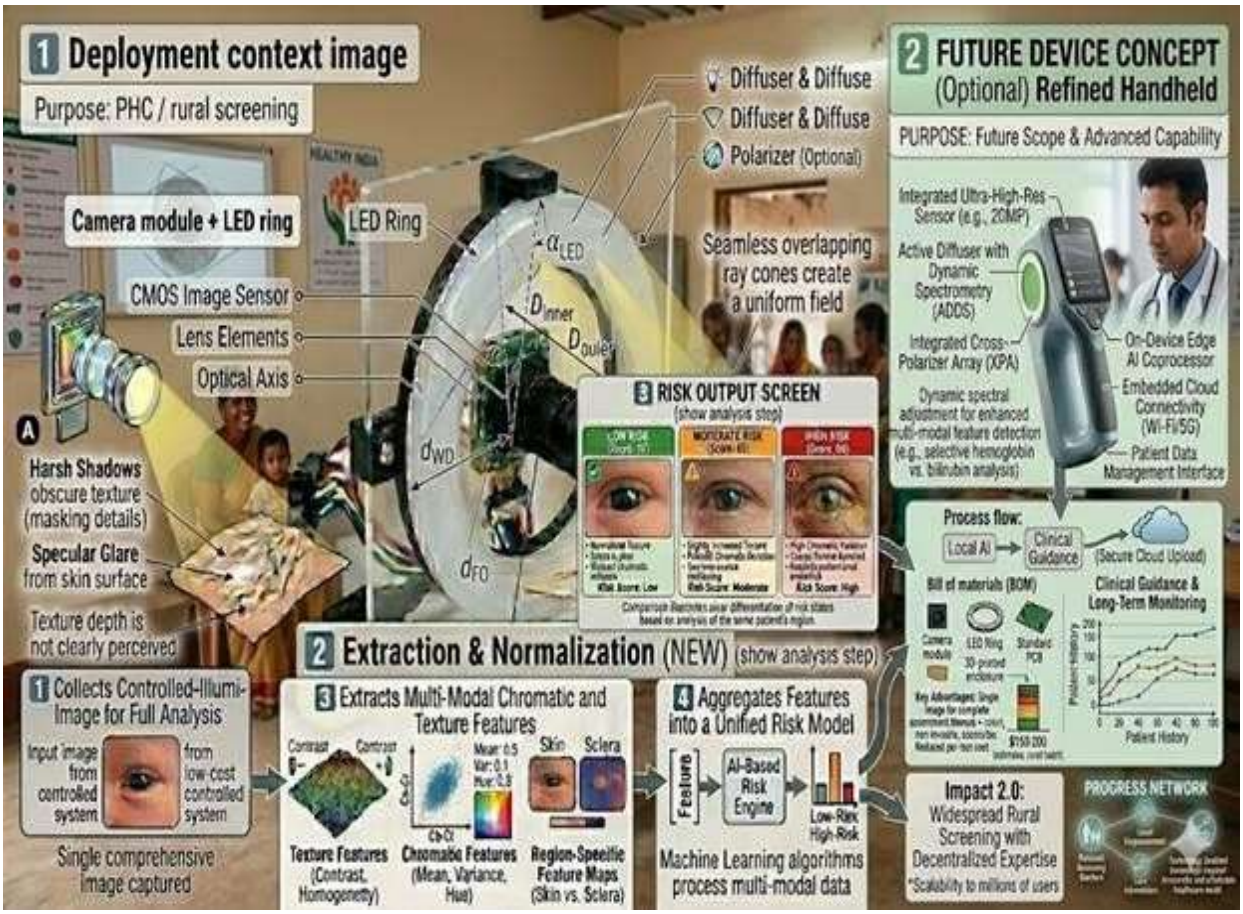
Machine Learning algorithms process multi-modal data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Specular Glare from skin surface
Texture depth is not clearly perceived
Diffuser & Diffuse
Diffuser & Diffuse
Polarizer (Optional)
Seamless overlapping ray cones create a uniform field
2 FUTURE DEVICE CONCEPT (Optional) Refined Handheld
PURPOSE: Future Scope & Advanced Capability
Integrated Ultra-High-Res Sensor (e.g., 20MP)
Active Diffuser with Dynamic Spectrometry (ADDS)
Integrated Cross-Polarizer Array (XPA)
Dynamic spectral adjustment for enhanced multi-modal feature detection (e.g., selective hemoglobin vs. bilirubin analysis)
On-Device Edge AI Coprocessor
Embedded Cloud Connectivity (Wi-Fi/5G)
Patient Data Management Interface
3 RISK OUTPUT SCREEN (show analysis step)
Process flow:
Local AI
Clinical Guidance
(Secure Cloud Upload)
Bill of materials (BOM)
Clinical Guidance & Long-Term Monitoring
Patient History
2 Extraction & Normalization (NEW) (show analysis step)
1 Collects Controlled-Illumi-Image for Full Analysis
Input image from controlled system
from low-cost controlled system
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Contrast
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4 Aggregates Features into a Unified Risk Model
Feature
AI-Based Risk Engine
Low-Risk High-Risk
Machine Learning algorithms process multi-modal data
Impact 2.0: Widespread Rural Screening with Decentralized Expertise
*Scalability to millions of users
PROGRESS NETWORK